Chapter 3

Empirical Tools of Education Economics

Chapter Outline

3.1 Descriptive Evidence and the Distinction Between Correlation and Causation

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3.3 Nonexperimental Methods

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Why Does College Enrollment Differ by Family Income?

If you compared students from wealthier family backgrounds with those from poorer ones, you would see a rather unsurprising pattern: Students from higher-resource

households are far more likely to go to college than their less-advantaged counterparts. These differences are large. Among students graduating from high school in the late 1990s and early 2000s, 29% from the lowest family income quartile went to college, while 80% from the top quartile did so[1]—a 51 percentage point gap.

Before we design a policy response to address this inequality in outcomes, it is important to understand *why* it exists. One prominent explanation is the role of costs. College is expensive in the United States, with tuition, fees, and room and board averaging more than $19,500 per year for a public four-year school and $43,900 for a private four-year school for academic year 2015–16. Even for two year colleges, which are considerably less expensive, costs could be expected to exceed $11,400.[2] Even for two-year colleges, which are considerably less expensive, costs exceed $11,000 per year on average. If the high price of a college education drives some of the enrollment gaps by family income, providing financial aid for low-income students may be a useful policy. Indeed, the federal government distributed more than $161 billion in financial aid in academic year 2014–15, and the core argument for offering such aid is to support the ability of students to pay for college.

Given the large differences in enrollment patterns by family income and the high cost of college, it is natural to ask whether financial aid would increase college enrollment among lower income students. However, we need to be

extremely careful about drawing such a conclusion from the negative correlation between family resources and college enrollment. For financial aid to increase college enrollment, it must be the case that a lack of financial resources among students at the time of the enrollment decision *causes* them not to enroll in college.

To see the potential problem with the causal interpretation of the data in this setting, consider the other ways students from low-resource backgrounds may differ from more advantaged students. Students whose parents have more money tend to have more education. Such parents may understand how to navigate the college enrollment process better, which could lead to higher attendance. Kids whose parents have fewer resources when they are of college age tend to have had fewer resources throughout their childhood. This can translate into lower levels of academic achievement by the time the students reach high school. Family resources thus are correlated with academic preparation for college, which should affect enrollment decisions. Isn't it possible that the differences in college attendance rates across students from lower- versus higher-income households are due to these other differences rather than the ability to pay for a college education? The short answer is yes; this is entirely possible.

As this example demonstrates, we need to find ways to establish the difference between two variables being correlated and an outcome being causally determined or

impacted by another variable. If the inability to pay college tuition is a main driver of the gap in college enrollment by family income, financial aid should be an effective policy tool to increase enrollment among low-income students. If these gaps largely reflect differences in parental education or early life educational investment, then financial aid will do little to address college enrollment gaps by family socioeconomic status. In effect, we would be deploying our resources in the wrong place because we misinterpreted the data. To justify a policy that provides financial support to low-income students for college enrollment, we need a way to disentangle the causal effect of the family's ability to finance tuition expenses on college enrollment from the other factors that may reduce collegiate investments for low-income students.

The empirical tools economists use to separate causation from correlation to inform education policy are the focus of this chapter. One of the core contributions of the social science of economics to the evaluation of education policies is the combination of *models* of behavior and *methods* for evaluating how education policies affect outcomes. These models and methods are not distinct: the models of behavior give us important predictions of the ways we expect individuals to react to specific policies and incentives.

This chapter discusses several of the tools economists and other social scientists use to determine whether a given policy causes education outcomes rather than simply being

correlated with those outcomes.[3] Our objective is to be able to answer questions like these:

- Do smaller classes lead to improvements in student achievement?
- Does providing more money to schools improve students' outcomes?
- Does completing a college degree lead to higher earnings?
- Do charter schools lead to gains in student achievement over regular public schools?
- How does teacher quality affect student academic outcomes?
- Does incentive pay for teachers improve students' academic achievement?

First, we consider what we can learn from descriptive measures as well as their fundamental limitations. Next, we introduce experiments in education as the basic scientific framework for understanding how education policies affect outcomes. While experiments in economics are relatively rare, they form a useful baseline with which to analyze policies in a nonexperimental setting. Then, we turn to the question of how we can estimate the impact of policies without the benefit of full experiments using **econometrics**. The concepts of **selection bias** and **omitted variables bias** are introduced; they are the fundamental problems to overcome when doing causal analysis. Finally, we discuss

three nonexperimental methods that are most commonly used by economists to overcome the biases associated with selection: difference-in-difference, instrumental variables, and regression discontinuity.

econometrics
The use of statistical techniques to measure relationships among variables in data.

selection bias
The bias that occurs because individuals choose whether they are part of the treatment or control group based on characteristics or preferences related to an outcome. This can lead to the characteristics of those in the treatment group being systematically different from those in the control group.

omitted variables bias
The bias that occurs when a variable is correlated with both the treatment and the outcome but is not included in the regression. This creates a bias in the estimate of the causal effect of the treatment, the sign of which depends on how the omitted variable is correlated with the treatment and with the outcome.

For policymakers to make good decisions about how to allocate resources, such as spending more (or less) money on particular policies, they must have a way to assess the causal effect of a given policy on outcomes of interest. Generating measurements and methods to assess these causal

relationships is often quite difficult. A central focus of the rest of this book is on providing an understanding of the ways economists have approached and attempted to solve these problems. The methods we describe in this chapter will be referenced with regularity throughout the rest of the book.

3.1 Descriptive Evidence and the Distinction Between Correlation and Causation

Researchers and policymakers have access to a wide array of data on the financing of education, the inputs to the education process, student achievement, and levels of educational attainment. Local, state, and federal governments collect a range of measures about schools, colleges, and students. Appendix A details many of the most widely used data sources in the economics of education.

One important use of these data is to *describe* outcomes. For example, we can record spending per student across districts in a state, the level of educational attainment among adults, or the distribution of test scores at a school. Typical descriptive measures include the mean, median, and percentiles of the distribution. The statistics provided in Chapters 1 and 2 are all examples of descriptive statistics. See the accompanying Toolbox for more details on descriptive statistics. Basic familiarity with descriptive statistics will help you read articles related to education

policy and approach this work as a critical consumer. Descriptive measures are important for accounting purposes —they tell us what the outcome is—but they don't tell us what the outcome would be if we changed policies.

TOOLBOX: Descriptive Statistics

Suppose we observe an educational outcome such as test scores for sixth-grade students in a particular school district. The test score for each individual (i) in the district is t_i. Assume there are N students in the district. The first and most common descriptive measure is the mean or expected value —$E(t)$, which is the sum of all individual values divided by the number of observations:

$$\mu t=E(t)=\sum i=1NtiN$$

Without any other information, the mean of the test score distribution is the best estimate of any individual's test score. We can compute the mean for subgroups of students, and we describe these measures as the conditional expectation. For example, we might be interested in the mean for girls or the mean at a particular elementary school and denote these conditional expectations $E[t \mid sex = girl]$ or $school = Lansing\ Elementary\ School]$.

Another measure of interest in describing the distribution of education data is the variance, or the spread of a distribution. A distribution with a large variance is going to

take on a much wider range of values than a distribution with a smaller variance. The variance of a distribution, often denoted var(t) or σ^2, is measured as the expected value of the square of the deviation of a variable from its mean. It is denoted as:

$$\mathrm{var}(t)=\sigma 2=E[(t-\mu t)2]=\Sigma i(ti-\mu t)2N-1$$

Other measures, such as percentile ranks and the median, are indicators of the dispersion of a distribution. The median is the value at which half of the cases are above it and half of the cases are below it.

When we have two or more variables of interest—such as class size (S) and test scores (t) or educational attainment (Ed) and earnings (Y)—we will often want to describe the extent to which these variables move together or in different directions. The covariance between variables S and t is defined as:

$$\mathrm{cov}(S,t)=\sigma S,t=E[(t-\mu t)(S-\mu S)]=\Sigma i(ti-\mu t)(Si-\mu S)N-1$$

Quick Hint: We divide by N – 1 in the variance and covariance formulas to adjust for degrees of freedom. The idea is that one must first use up one observation to calculate the mean before calculating the variance. There are N – 1 free observations once one calculates the mean. To calculate the covariance, one must first calculate the mean, which also uses up one observation. So there are only N – 1 free observations. See Wooldridge (2009) for a

more detailed discussion of degrees of freedom adjustments.

Intuitively, the covariance will be positive in cases where observing t greater than the mean of t is accompanied by observing a value of S greater than the mean of S. If on average t is greater than the mean when S is less than the mean, there is a negative covariance between the two variables. When knowing that one variable is greater than (or less than) its mean on average provides no systematic information about the likelihood that another variable is greater than (or less than) its mean, and the associated covariance is zero.

Given that covariance measures are dependent on the units of observation (multiplying a variable by 100 will change the measure of the covariance), it is common to use the measure of correlation as a standardized description of the association between two variables. The **correlation** measure, which will always have values between −1 and 1, is defined as:

correlation
The extent to which variables move together in the data.

$$\rho S,t=\mathrm{cov}(S,t)\mathrm{var}(S)\mathrm{var}(t)=\sigma S,t\sigma S\sigma t$$

Descriptively, we can measure how variables move together, the degree of correlation. For example, we can measure the association between finishing college and earnings. On average, college graduates have higher earnings than those who stop formal schooling at high school, so education and earnings are positively correlated. Similarly, we would likely find that educational attainment and arrest rates are negatively correlated.

The most important point to take from the observation of the relationship between two variables such as education and earnings is that a correlation does not necessarily imply a **causal link**.[4] The observation of a positive link between education and earnings need not imply that increasing an individual's educational attainment would lead to an increase in earnings. It is often the case that we can describe the relationship between two variables (the correlation between *X* and *Y*) without demonstrating a causal relationship (the effect of changing *X* on outcome *Y*).

causal link
between two variables; altering one variable leads directly to a change in the other variable. That is, a change in one variable results in a change in another.

Knowing that two measures are correlated tells us little about the causal effect of changing one policy variable on an educational outcome. Suppose we are interested in the link

between class size and student achievement. Looking across students, we may see that those in smaller classes have higher achievement. Such a descriptive result does not reveal how changing class size would affect achievement because we don't know *why* class size differs among students. It may be that students with higher (or lower) achievement are systematically assigned to smaller classes. Alternatively, it may be that schools with more resources across a number of dimensions, including teacher quality, also have relatively small classes, so it is not class size per se but some other third factor that is causing the higher achievement in schools with smaller classes.

For the purposes of education policy, we need to answer causal questions, such as how changing an individual's educational attainment would affect earnings or how reducing class size would affect a student's achievement. Suppose we want to know whether completing another year of high school is likely to make *any given student* earn more than if he were to drop out. Imagine a hypothetical situation in which we can "treat" an individual with more schooling or not. We want to know how the treatment will affect earnings. The earnings outcome is designated as Y, and we use the subscript T (the "treatment") for an additional year of schooling and C (the "control") for unchanged schooling. To answer this question for a single student, whom we will call Larry, we would like to know:

$Y_T(Larry)$ = Larry's wage if he finishes one more year of high school
$Y_C(Larry)$ = Larry's wage if he doesn't finish another year of high school

We want to measure the earnings gain from attending another year of high school which is in Larry's earnings in these two states:

$$\textbf{Treatment effect} = Y_T(Larry) - Y_C(Larry)$$

treatment effect
The causal effect of the treatment on a specific outcome.

You may have noticed that we have a big problem in performing this calculation: there is only one Larry. He could not have concurrently finished another year of high school and not done so. Put another way, the problem is that we observe only one outcome for each person. We do not observe the **counterfactual**, or what would have happened in the absence of the additional educational attainment. In the absence of observing Larry in both education states, we would like to observe someone who is identical to Larry in every way *except* that he took a different schooling path. We thus want to use another similar individual to measure Larry's counterfactual earnings outcome.

counterfactual
What would have happened to an individual in the absence of the treatment.

Deep Dive: Why Is It Important to Distinguish Correlation from Causation?

The distinction between causation and correlation is fundamental in the design of policy. To illustrate how correlations fail to inform policy, it is instructive to consider some examples that border on the absurd.

1. Does weight gain reduce your intelligence?

A study done in France on 2,200 adults examines how changes in weight gain are associated with changes in performance on a cognitive ability test. They find that those who gain weight have reduced cognitive ability, which they interpret as a causal effect of weight gain on intelligence (Goswami, 2006). If true, this would seem to argue for policies that lead to weight loss among adults, because the costs of lowered intelligence are high. But we need to be very careful in assessing the claims of causality in this study. One way to think about this problem is to ask, 'Why do people gain weight?' It could be that an adverse life event makes them less motivated, which would also show up on a

test score. A similar story holds for any adverse health shock. Alternatively, what if those who lose intelligence gain weight? While the correlations found in this research are valid, the causal claims likely are not.

2. Does reading 50 Shades of Grey *make you less healthy?*

The novel *50 Shades of Grey* was a bestselling erotic romance novel. In a research study comparing outcomes among young women ages 18–24 who read the novel to those who did not, Bonomi et al. (2014) found evidence that those who read this book were more likely to binge drink, to be in abusive relationships, and to have more sexual partners. Is the conclusion that there would be benefits from banning this book? This depends on whether these outcomes are caused by reading *50 Shades of Grey*. Rather than being causal, it is highly likely that women who drink more, who have more sexual partners, and who are in abusive relationships are more likely to read this book, in which these behaviors all are prevalent.

3. Does smoking pot lead to less domestic violence?

Researchers at the University of Buffalo found evidence that couples who smoke marijuana had fewer incidences of domestic violence (Smith et al., 2014). It could be that smoking pot leads to less domestic violence, which might argue in favor of relaxing restrictions on this drug. An alternative and highly plausible explanation for these findings is that couples who smoke pot are less likely to engage in domestic violence in the first place. That is,

selection into smoking pot drives this correlation. In such a case, legalizing marijuana would have little effect on domestic violence rates.

4. Do big weddings lead to stable marriages?

The Relationship Development Study, conducted by psychology researchers at the University of Denver, tracks 418 initially single people between the ages of 18 and 40 over time. When these individuals got married, the researchers found that the marriages were more likely to last if the wedding had at least 150 people in attendance (Kaplan, 2014). Does this mean you should spend the money for a big wedding? You may want to do so for the fun of it, but you should be careful in using these findings to justify such an expense. It is probable that couples with large weddings have larger support networks, higher incomes, and stronger family bonds. These are more likely to be the mechanisms through which marriages remain more stable. Big weddings are a result of these mechanisms, so increasing the size of your wedding is not likely to lead to a better marriage.

5. Does drinking coffee while pregnant lead to miscarriage?

Researchers at Kaiser Permanente examined the caffeine intake of 1,063 pregnant Kaiser patients in the San Francisco Bay Area (Wang, Odouli, & Li, 2008). They found that women who consumed over 200 mg of caffeine per day (about 2 cups of coffee) had twice the risk of miscarriage as those who consumed none. It is tempting to conclude from

this study that one should not consume caffeine while pregnant. Although many women do make this choice, the findings of this study do not support such a policy. We must ask why some women consume more coffee than others while pregnant. It is highly likely that coffee consumption patterns are related to outside stress from family pressures or work. It thus is quite plausible that these stress factors drive the miscarriage result, not the amount of coffee one drinks.

Our challenge is to find a valid counterfactual. This is very difficult because we can't observe the same individual in different states of the world. One approach is to use random assignment to generate comparison groups that are similar on average. We also consider other ways in which the nature of policy design may generate similar groups that did and did not receive the education intervention. If we can measure average outcomes for groups that are identical in all respects except for receipt of an education intervention, we can capture an estimate of the causal effect of the treatment. This is our challenge.

3.2 Randomized Control Trials: The Experimental Ideal

Experiments, or **randomized control trials (RCTs)**, are often viewed as the gold standard for measuring causal effects. In a randomized control trial or experimental design, one group of individuals (or other units, like classrooms or schools) is assigned to a treatment status to receive some intervention, while others are assigned to control status and are not subject to the intervention. The idea in an experiment is to separate participants into a treatment group and a control group, using random assignment. The treatment might be receiving a new medication in the medical sciences or, in our context, participating in additional schooling. The key to the design is that individuals are randomly assigned to the treatment status. This ensures that, on average, outcomes would be identical in the two groups in the absence of the treatment.

randomized controlled trial (RCT)

An experiment in which people are randomly assigned to the treatment and control groups. On average, this makes

the two groups identical but for receiving the treatment.

Participants in the experiment are necessarily not perfectly identical, but because we have randomly assigned participants to treatment and control groups, there should be no differences, on average, in baseline characteristics between the two groups. Any differences between the outcomes of the treatment and control groups at the end of the experiment must be due to the treatment: The control group is an accurate measure of the counterfactual outcome for the treated group.

Let's think for a moment about what random assignment implies about treatment and control groups. We would expect those assigned to the treatment and control groups to have similar observable characteristics. If we started with 100 students and randomly assigned those students to be in the treatment group or control group, we would expect that the proportion female, average prior test scores, and average family income would be similar across treatment and control groups. We can observe whether this condition holds, and we would expect these measures to be more similar as we increased the total number of students in our study. Importantly, with random assignment we also expect treatment and control groups to be similar on average in dimensions that are difficult to measure, such as motivation or determination.

The measurement of the causal effect in the context of a

randomized control experiment is straightforward: One need only compare outcomes for the individuals in the treatment group to the outcomes of individuals in the control group. If our outcome is a measure of student achievement, the causal effect of an RCT intervention like class size reduction is simply the difference in average test scores between the treatment group and the control group. Table 3.1 illustrates the simple comparison needed to measure a treatment effect with a random assignment experiment.

Table 3.1 Measuring Treatment Effects in an Experimental Context

Group (randomly assigned)	Mean Outcome
Treatment	A
Control	B
Treatment effect	A − B

In an experimental setting, the treatment effect can be calculated as the difference in outcomes across those randomly assigned to the treatment versus those randomly assigned to the control group (A − B).

Randomized trials have long been used in medicine to test the effectiveness of new treatments and as a tool for discovery in the life sciences and the physical sciences. Historically, experiments have been far less common in education, though certainly not unheard of. Perhaps the most widely cited experiment in education is the 1962 Perry preschool project involving the random assignment of low-

income children from Ypsilanti, Michigan, to a treatment that included intensive preschool and home visits. This study found generally positive long-term gains associated with early education intervention. Until recently, experiments like this in education have been few and far between. Indeed, in 2000 only one of the 84 projects that were part of the Department of Education's annual plan involved a randomized trial (Angrist, 2004), but the standards of evidence have changed dramatically in education research in recent years. One factor in this shift is the 2002 Education Sciences Reform Act, which emphasizes "scientifically based research" and explicitly identifies randomized control trials as the gold standard in research design.

Table 3.2 describes some of the most widely known experiments related to education policy. We will discuss the results of many of these experiments in later chapters in the context of particular policy questions. While some of these experiments have been conducted by economists, others have been designed by psychologists and social scientists from other disciplines. Education experiments include random assignment to various preschool settings, vouchers for elementary schooling, particular pedagogical tools such as computer-based instruction, and access to assistance with completion of financial aid forms and college application information.

The reason well-designed experiments are so persuasive is that they address the fundamental problem of selection. In

a nonrandomized setting, agents are allowed to choose a treatment, such as whether to get another year of education or whether to pursue an intensive math curriculum. Economic theory tells us that individuals will choose the activity that will maximize their utility—that is, choose the activity that will make them the best off. As a result, those who take the treatment are likely to be systematically different from those who do not. So at the end of the day, differences associated with the program or intervention also would capture systematic differences in individual characteristics correlated with the outcome. When an estimation method produces a result that differs systematically from the causal effect of interest, we say that method is a **biased estimator**. *Selection bias* is a particular form of bias driven by that fact that individuals can choose whether they are in the treatment or not. Since economists tend to focus on how people make choices when resources are limited, selection bias is the most prevalent problem addressed by economists engaging in causal analysis. The beauty of random assignment in the RCT is that selection bias should be zero by design.

biased estimator

A method of estimating causal effects is biased if, on average, the resulting estimate differs from the true causal effect.

Table 3.2 Examples of Random Assignment Experiments in Education

Experiments in Education	
Experiment	**Basic Strategy**
Project Star	Random assignment of students in Tennessee to small and large classes in grades K–3
Perry preschool	Random assignment of poor children to an intensive preschool program
Computer instruction: Fast ForWord	Evaluation of program designed to improve language and reading skills
H&R Block financial aid information	Random assignment of low-income students and parents to receive information about financial aid and assistance with financial aid forms
Private school tuition vouchers	Experiments in Milwaukee and New York City that provide vouchers to low-income youth to attend private schools
Expanding College Opportunities Project	Provided information about college application strategies, college quality, and application fee vouchers to a randomly selected set of low-income, high-achieving high school students

The ideal experimental design that forms the basis for most laboratory and pharmaceutical experiments not only employs random assignment of the treatment but is also *double blind*—neither the participants nor those running the experiment know which group is treated. It is very difficult in education to conduct experiments that use this ideal design because both the researchers and participants often know who is receiving the treatment. As a result, randomized field

experiments in education are often more challenging to design and study than typical laboratory experiments.

We say a research design has **internal validity** if the estimated causal effect of the treatment is unbiased. Threats to internal validity can come from several sources:

internal validity
The extent to which the estimated causal effect of the treatment is unbiased.

- Differential attrition between treatment and control groups
- Sampling error[5]
- Knowledge of the treatment (or control) status, which independently affects behavior

Typically, a treatment is an intervention that is desired by potential participants. For example, a class size reduction experiment involves a treatment—smaller classes—that is likely to be of value to children and parents. As a result, parents assigned to the control group may get angry and leave the school. In this setting, parents know whether their child is in a smaller class, which can lead to differential attrition. If the highest-achieving students in the control group (or, more generally, students with different achievement levels) drop out of the sample, the estimate of the causal effect of the program will be biased. This occurs because the control

group is now lower-achieving than the treated group: The higher-achieving control group students left the school. In social sciences, we cannot compel people to comply with their assigned treatment status or to stay in an experiment, which can cause many problems for estimating causal relationships.

Another problem is sampling error: If the experiment has few people in it, the treatment and control groups may differ for random reasons. That is, the groups differ by accident. Including a larger number of people in the experiment reduces this problem, although it also increases costs.

The inability to perform double-blind experiments in education policy also leads to the possibility that the treatment assignment itself could independently affect participants' behavior. One example of this problem is called the **Hawthorne effect**, which refers to the phenomenon that when people know they are in an experiment to demonstrate a particular effect, they are likely to behave in a way to make that effect occur. For example, teachers who know they are in a class size experiment may behave in accordance with the study's hypothesis: teachers of small classes will perform better than those with larger classes. This response is not due to the effects of class size per se but is due to the knowledge of being in the experiment.

Hawthorne effect

What happens when people know they are part of an

experiment and behave in a way that is more likely to make the hypothesis being tested seem true.

A final concern with randomized controlled trials is that even when they lead to internally valid results, the estimated effects may lack **external validity**. External validity pertains to whether we can generalize the results to other settings. For example, results from a study in Iowa may not necessarily apply in California if students' and institutions' characteristics are vastly different. RCTs with very small samples also make the results unlikely to be externally valid, as the small group of people studied are unlikely to be representative of state or national populations.

external validity
The extent to which we can generalize results from an empirical study to other settings.

External validity can be a particular concern with many education interventions because RCTs often are conducted among a group of participants who volunteer for the experiment. RCTs can tell us what the effect of an education treatment is among those who sign up for the experiment, but strong assumptions are needed to apply this estimate to those who do not sign up.[6] If we want to scale up an intervention to a broader population, it is critical to understand its effect on those who chose not to be a part of the experiment. External

validity therefore is a critical issue to consider when using results from RCTs to design education policies.

3.3 Nonexperimental Methods

Many important questions in education policy cannot be examined with randomized controlled trials. Substantial time lags to observe results combined with high costs are one deterrent to the randomized control approach. In addition, in a number of circumstances it may not be logistically or ethically feasible to employ a random assignment design. For example, assigning students at random to receive admission to selective universities might result in considerable objections to implementation. Even the proposed experiment with Larry to assign him an extra year of education is unlikely to be logistically feasible because it is not possible to "force" completion of a year of school, not to mention morally acceptable. While experiments that are short-term and do not involve large time commitments are straightforward, very resource-intensive experiments are typically extremely expensive while also raising ethical questions when children are participants. As a result, economists have turned to nonexperimental methods that in some circumstances can also overcome selection biases. These methods use observed variation in outcomes that is plausibly unrelated to both observed and unobserved

individual characteristics to estimate causation. In short, economists must mimic experiments by using naturally occurring variation.

To solve vexing questions like estimating the causal effect of education on earnings, we must build a rich methodological toolkit and think creatively about how to use the data at hand to statistically isolate causal effects when we cannot conduct an RCT. The main workhorse of nonexperimental economics is *multivariate regression*. This is the tool most used to estimate causal effects, but as we shall see, using multivariate regression is *not* sufficient for isolating causality. Multivariate regression is a tool, and like all tools, it must be used correctly to get the desired result. While we will leave many of the more advanced methodological tools to a course in econometrics, it is useful for every student to have a baseline understanding of regression analysis and statistical inference.

> **Quick Hint:** By desired result, we do not mean the specific result of the study. Rather, the desired result is the causal estimate that overcomes the problem of selection bias, regardless of whether this is the result any constituency would "like" to see.

Regression Analysis

If we want to know how another year of education affects earnings, we might start with a simple plot with observed

education on the horizontal axis and observed earnings on the vertical axis. What would we expect? We would likely expect to find that individuals with higher levels of education had, on average, higher levels of earnings. We would not expect to see a perfect correlation, with all combinations of education and earnings on a single line, as there are many determinants of earnings beyond education. Figure 3.1 shows this type of positive relationship between observed educational attainment and earnings.

To fix terms, we call education (*Ed*) the **independent variable (explanatory variable)** and earnings (*Y*) the **dependent variable**, or outcome variable. With a positive relationship between education and earnings, knowing an individual's level of education is going to help predict that person's earnings. Assuming that the link between education and earnings is linear in the sense that each year of education predicts a constant increase in earnings, we could describe the data as:

independent variable (explanatory variable)
The variable in a regression used to describe the dependent variable or outcome of interest.

dependent variable
In a regression, the variable we are seeking to explain with the independent or explanatory variables. It is the outcome of interest in a regression.

$$Y_i = \alpha + \beta Ed_i + \varepsilon_i$$

where the subscript *i* designates individual observations; α is the intercept, or the amount a person with zero years of education could expect to earn; β is the slope; and ε represents other determinants of individual earnings. Regression analysis is a statistical technique to pick the α and β that best fit the data; it is common to designate estimates of these parameters as σ^ and β^. We can describe the relationship between education and earnings predicted by a regression as the line Y=α^+β^Ed, as shown by the red line in Figure 3.1, where β^ is the slope of the line and σ^ is the *y-intercept*. This is an example of a *bivariate regression*, because there are only two variables, earnings (dependent variable) and *Ed* (independent variable).

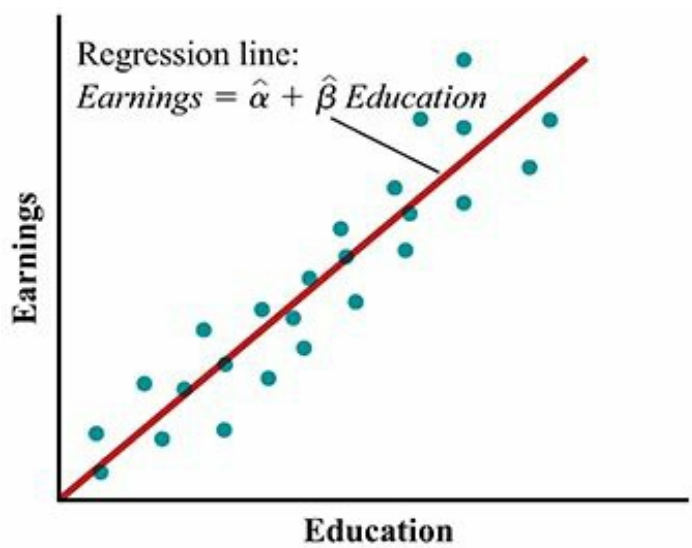

Figure 3.1
Lovenheim/Turner, *Economics of Education*, 1e,
© 2018 Worth Publishers

Figure 3.1 Illustrating Basic Regression

Analysis Using Education and Earnings

Each dot represents a combination of education and earnings for a different individual. The regression line is the line that best fits these data: The slope of the regression line is the regression coefficient on education in the bivariate regression of earnings on education, $\hat{\beta}$.

Quick Hint: In statistics, we distinguish between the true values of the parameter that we can never observe and our estimates of those parameters that we generate using data. The true, unobserved parameters are the population parameters, and one can think of regressions as trying to estimate these unknown population parameters.

Suppose that we had randomly assigned a large number of individuals to different levels of education. Our estimate of $\hat{\beta}$ would be equal to the experimental estimate, and comparing the earnings expected with $Ed = 1$ and $Ed = 0$, produces:

$$E[Y_i|Ed=1]-E[Y_i|Ed=0]=\hat{\alpha}+\hat{\beta}-\hat{\alpha}=\hat{\beta}$$

In this context, a regression estimate is just a particular way of presenting an experimental result.

Quick Hint: E [Y | X] is called a conditional expectation function. It shows the expected value (i.e., average) of Y

for each value of X. In this setting, E [Y | Ed] shows average earnings for each level of education. The parameter β^ therefore shows how average wages change when education changes by one year, which is the slope of the line shown in Figure 3.1.

Now, what if we can't run an experiment but are able to observe a large number of people with different levels of education? We still can compute an estimate of β^, but we should be very cautious about interpreting this measure of the association between education and earnings as the causal impact of education on earnings. The problem is that individuals who choose to get more education are likely to differ from those who choose lower levels in ways that are also related to earnings. The counterfactual earnings for the high-education group is unlikely to be the earnings that we observe for the low-education group. Suppose that individuals with higher underlying earning potential tend to choose higher levels of educational attainment. The bivariate correlation between earnings and education then will reflect this ability difference as well as the effect of education on earnings.

Figure 3.2 illustrates this problem, which is a canonical example of selection bias. We have plotted education attainment and earnings for high- and low-ability students. Students' ability is measured by SAT scores in this example, which we do for illustrative purposes. As the figure shows, it is true that those with more education earn more, but they

also have higher SAT scores on average. If there is an independent return to this ability in the labor market, the bivariate regression will not allow us to separate the effect of SAT scores from the effect of education on earnings. As panel A of the figure shows, the differences in earnings by education capture both the causal effect of education and the differences generated by selection of higher-ability people into receiving more education.

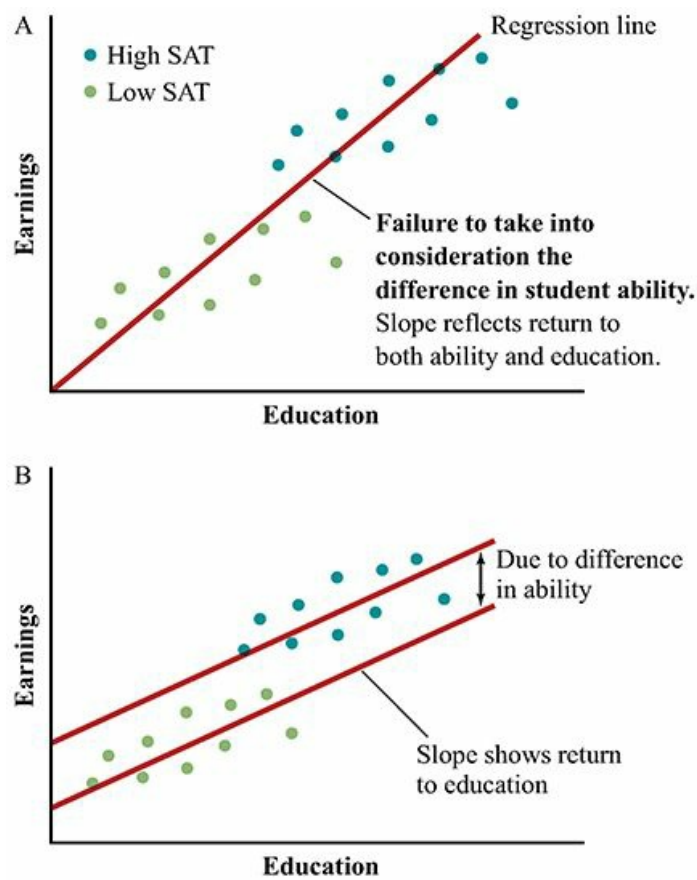

Figure 3.2
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Figure 3.2 Ability Bias in Regression Analysis

Both panels show a plot of education attainment and earnings for high- and low-ability students,

with students' ability measured by SAT scores. **A.** Differences in individual ability correlated with education. We do not control for ability, and as a result the estimated regression line reflects both the return to ability and the return to education. **B.** Including an ability measure in the regression. After we control for ability, the regression slopes become flatter, as they now reflect only the return to education.

Quick Hint: The SAT is a college entrance examination that is accepted by most schools in the United States. Alternatively (or additionally), many students take the ACT. The ACT is another widely used college entrance examination that is most popular in noncoastal states.

From an econometric standpoint, ability here is an **omitted variable**, which leads to an omitted variable bias. An omitted variable bias occurs when a variable is not included in the regression but is correlated *both* with the treatment and the outcome. The Toolbox has a more detailed derivation and discussion of omitted variables bias in regression models. Omitted variables bias and selection bias are basically the same thing: There is some reason why the treated group is systematically different from the control group. In essence, selection causes omitted variables bias. In our example of education and earnings, the omitted variable bias is caused by the selection of higher-achieving students

into receiving more education.

omitted variables

In a regression, any variables that are correlated with both the treatment and the outcome that are not included in the estimation.

Toolbox: Omitted Variables Bias

In this toolbox, we derive the formula for omitted variables bias and discuss how one can sign this bias when thinking about the influence of a potentially omitted variable in an empirical study. Let's begin by considering a general regression with two independent variables:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

where Y is the dependent variable, X_1 and X_2 are the independent variables, and ε is the error term. Assume that this model is the true (i.e., population) model; it correctly describes the relationship among the three variables. Consider what happens if we have a data set that does not include X_2. We thus estimate a bivariate regression of Y on X_1:

$$Y = \gamma_0 + \gamma_1 X_1 + \mu$$

Recall that the definition of bias is that our estimate must on average differ from the truth. This means that, on average, our estimate γ^1 must equal β_1 if our estimate is to be unbiased. Under what conditions will this occur? The expected value of the bivariate linear regression estimator can be written as:

$$E(\gamma\hat{}1)=cov(Y,X1)var(X1)$$

where cov(Y, X_1) is the covariance between Y and X_1 and var(X_1) is the variance of X_1. If we substitute $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$ for Y in this expression, since by assumption this equality holds, we get:

$$E(\gamma\hat{}1)=cov(\beta 0+\beta 1X1+\beta 2X2+\epsilon,X1)var(X1)=cov(\beta 0,X1)var($$

In the expression on the second line, the first piece is zero, since β_0 is a constant and the covariance between any variable and a constant is zero. The second piece reduces to β_1, as cov(X_1, X_1) = var(X_1) by definition. The final piece is also zero, since by assumption of the linear regression model, the error term is uncorrelated with each independent variable. We are left with the following expression for the expected value of β^1:

$$E(\gamma\hat{}1)=\beta 1+\beta 2cov(X1,X2)var(X1)\equiv\beta 1+\beta 2\delta$$

The bias in this formula is given by $\beta_2\delta$. This bias has two parts. The first part is the coefficient from the true model on X_2. The sign of the bias therefore depends on how X_2 and Y are correlated. The second piece $\delta\beta$ is the regression coefficient from a bivariate regression of X_2 on X_1. We cannot actually estimate this regression, since our data do not include X_2, but the sign of this coefficient is driven by the correlation between X_2 and X_1. This is a very useful piece of information.

The sign of the bias when we cannot control for X_2 depends on the signs of β_2 and δ. When they have the same sign, the bias is positive, and when they have different signs, the bias is negative. Let's think through a couple of examples. First, imagine Y is yearly earnings, X_1 is education, and X_2 is ability. If we estimate a regression of earnings on education (omitting ability), what will be the sign of the omitted variables bias? The sign of β_2 is determined by the correlation between ability and earnings, which almost certainly is positive: All else equal (including educational attainment), higher-ability people earn more. The sign of δ is determined by the correlation between education and ability, which also is likely to be positive: Higher-ability people get more education—see Chapters 4 and 6. The omitted variables bias will be positive: Education will look more important than it is in determining earnings because we have not controlled for ability.

Consider the potential for bias if we estimated a bivariate regression of test scores at the school level on average class size in the school. Depending on the circumstances, the bias could go either direction. On one hand, suppose students in schools with higher-income parents are in schools with smaller class sizes. It is likely that parental income is positively correlated with test scores ($\beta_2 > 0$). Since parents with more income tend to sort into schools with higher resources, we would expect parental education to be negatively correlated with class sizes ($\delta < 0$). The omitted variables bias therefore is negative: Class sizes look as if they have a larger negative effect on test scores in the bivariate regression than they actually do because we have failed to account for the confounding influence of parental education.

On the other hand, resource distribution may be *compensatory*, such that more educational resources are given to more disadvantaged students. In such a circumstance, parental income would positively covary with class size ($\delta > 0$). While we expect the true effect of class size on achievement to be negative ($\beta_1 < 0$), the omission of parental resources in the compensatory case will bias the estimated effect upward or toward zero as ($\beta_2 > 0$, $\delta > 0$). This makes class sizes appear less linked to test scores than they actually are. To predict the sign of the omitted variable bias, we need to know something about why different students experience different class sizes.

The omitted variables bias formula gives us a framework for thinking about what belongs in a regression. If what we care about is estimating the causal effect of X_1 on Y, then we need to control for all variables that are correlated with *both* X_1 and Y. In the education–earnings regression, we need to control for all variables that are correlated with both education and earnings; a variable that is correlated with only one of these will not produce a bias. Similarly, for the test score–class size regression we need to control for all variables correlated both with class sizes and with test scores.

Since in both of these examples the set of variables that are correlated with the treatment and outcome is large and because many of them are hard to measure (e.g., motivation and preferences for education), it is rare that a study can credibly overcome problems associated with omitted variables bias simply by controlling for observable characteristics of students and/or schools. Rather, to avoid biases from omitted variables, we typically look for sources of variation in the treatment that will be uncorrelated with these characteristics.

What can we do to address selection bias? There are two main solutions, each of which can be implemented in several ways:

Include in the regression control variables that account for

the ways in which the treatment and control groups are different.

Find a source of variation in the treatment variable that is unrelated to underlying differences between those who are treated and untreated. We need variation in the treatment that is generated by luck or randomness, not by individual preferences and abilities.

Every nonexperimental paper we will discuss in this book uses one of these two methods to establish causality.

The first approach can be thought of as attempting to address the selection problem by making comparisons among treated and untreated individuals who are similar in all observable dimensions that might also affect our outcome. For example, rather than comparing individuals with different levels of high school achievement, we might make our comparison among individuals with the same level of high school achievement. Among individuals with the same level of high school achievement, we may be able to assume that differences in educational attainment are plausibly just random, or unrelated to eventual outcomes. If this assumption is correct, then we can measure the effect of our education variable on our outcome of interest by controlling for the observed characteristics of students that make treatment and control students different.

Returning to the regression framework for our education and earnings example, we could control or condition on high school achievement (SAT scores). In essence, we would

measure the causal effect as the difference in outcomes for groups with the same level of achievement:

$$E\,[Y_i \mid Ed = 1, SAT = High] - E[Y|Ed = 0, SAT = High]$$

or

$$E\,[Y_i \mid Ed = 1, SAT = Low] - E[Y|Ed = 0, SAT = Low].$$

We also could use a regression framework to specify this relationship. If the education treatment has the same effect on low-achieving and high-achieving students, we can specify the relationship as

$$Y_i = \alpha + \beta Ed_i + \delta Ability_i + \varepsilon_i$$

and estimate the parameters α^, β^, and δ^. This is an example of a *multivariate regression*, as there are multiple independent variables in the model. What we have done is to specify a different intercept in earnings for high-ability students relative to low-ability students, which leads to an unbiased estimate of β^ so long as there are no other omitted determinants of education that also affect earnings. This scenario is shown in panel B of Figure 3.2. The relationship between education and earnings now is much smaller, although it still is positive. Because we did not control for ability in panel A, our estimate of the effect of education on earnings was biased upward: Education looked more important for earnings than it actually is because we were

conflating the effects of education and ability.

In panel B of Figure 3.2, the effect of education on earnings is the same for the high- and low-SAT groups. This need not be the case. In general, we do not require that treatment effects are the same for individuals with different characteristics. Variation in treatment effects with individual characteristics is called **heterogeneous treatment effects**. Multiple regression techniques can handle these cases simply by including interactions between the treatment and the relevant characteristic.

heterogeneous treatment effects

When the treatment has different effects on those with different background characteristics. For example, financial aid policies likely have heterogeneous effects on low- versus high-income students.

Multivariate regression analysis can be a very important and powerful tool. It allows analysts to control for observed differences among individuals and schools when trying to ascertain how education policies or interventions affect student outcomes. When using these methods, it is critical to recognize that the set of characteristics we can control for is limited by the data. We are able to condition only on the *observed* differences that may be related to the intervention and the outcome. To the extent that there are important *unobserved* differences among individuals or schools that

are related to both the policy and the outcome, the regression estimates will be biased. Since many aspects of education decisions are based on factors we do not observe, like motivation and preferences, it is rare that simply controlling for observed differences across individuals or schools is sufficient to establish causality.

Because of these limitations, most economics research in education seeks to use the second method: Find a source of variation in the treatment that is uncorrelated with differences across individuals or institutions. This is much easier said than done. Economists have developed a number of tools and strategies to make comparisons in ways that focus on differences in education treatments generated only by processes that are, in effect, similar to random assignment. For this reason, we often refer to these approaches as **natural experiments (quasi-experimental designs)** to capture the point that we are trying to think as if we could run an experiment. Knowledge of policy changes and how rules for determining education requirements work are the main ways in which economists approximate experiments using observational data.

natural experiments (quasi-experimental designs)
Use of variation in treatment exposure determined by nature or changes in policy that are outside the control of the researcher but nevertheless approximate random assignment.

Policy Changes and the Difference-in-Difference Approach

The Difference-in-Difference Approach

When governments change policies related to funding for education or requirements for participation in schooling, they create opportunities to evaluate the effects of these policies. Particularly when policy changes are not fully anticipated by students, parents, and schools, the measurement of outcomes before and after the policy change can be a powerful way to assess a program's effects. Examples include the imposition of compulsory attendance laws, which make some students stay in school longer than they otherwise would, or the introduction of a financial aid program that reduces the price of college.

The easiest way to proceed in such situations is to look at how students' outcomes change after the policy is implemented relative to before. For example, if a state increases its compulsory schooling age from 16 to 17, we can see how the educational attainment of 16-year-olds changes over time in the state. We would expect the pool of 16-year-olds in a given state in two consecutive years to be very similar along observed dimensions (e.g., test scores, gender, race) as well as along unobserved dimensions (e.g.,

motivation, unmeasured skills), which satisfies one requirement for unbiased estimation of our policy effect. With this change in the compulsory attendance law observed at the state level, comparisons of attainment and earnings before and after the change provide a measure of the policy effect.

Looking at outcomes before and after may still leave some concerns about whether the before and after cohorts would have had the same outcomes in the absence of the policy change: Is the before cohort an accurate counterfactual for the after cohort? In many ways we should think so. However, what if there is an economic shock (such as a recession) in the same year as the law change? What if educational attainment among 16-year-olds is increasing over time for reasons having nothing to do with the law change? In both cases, we will be unable to disentangle these unobserved shocks and secular trends from the effect of the policy itself.

To account for these other factors, we can look at outcomes over this period for a group that is otherwise similar but that was not subjected to the policy change. As long as the national economic shock or the trends in 16-year-olds' enrollment are similar in the unaffected and affected areas, we can use changes among those in the unaffected areas to control for these confounding factors. In effect, we can "difference" or subtract the change among the untreated or control group from the change observed among the treated group. This is the idea of the *difference-in-difference*

approach to causal estimation.

What we need to do is to establish a measure of the counterfactual change in the absence of the policy shift. Let's make the difference-in-difference approach concrete with an example. In 1993, the state of Georgia instituted a merit aid program, called HOPE, providing essentially free tuition at Georgia public postsecondary schools to Georgia high school graduates with a high school GPA above 3.5. Our policy question: Did this program increase college enrollment among Georgia high school graduates? We observe that enrollment rates in public universities among Georgia high school graduates increased, but a careful analyst would question whether this result was caused by the introduction of the scholarship program. College enrollment rates at these schools might have increased in the absence of this aid program, as this was a period of relatively robust economic growth in Georgia and other states in the southern United States.

We can address this concern by thinking about the enrollment change in states that border Georgia as providing our counterfactual change in enrollment. Figure 3.3 illustrates the observed trend in Georgia as well as the observed trend in bordering states. As you can see, there is a level difference between the two: Enrollment in the bordering states is higher in both periods. This alone is not a problem, as the difference-in-difference method controls for fixed differences across treatment and control individuals, states,

or schools. The difference-in-difference estimate, then, is the posttreatment difference between Georgia and the bordering states minus the pretreatment difference between the two groups. This is shown in the figure as $(D - B) - (C - A)$. Alternatively, one can calculate this as the change in Georgia minus the change outside of Georgia, which is $(D - C) - (B - A)$.

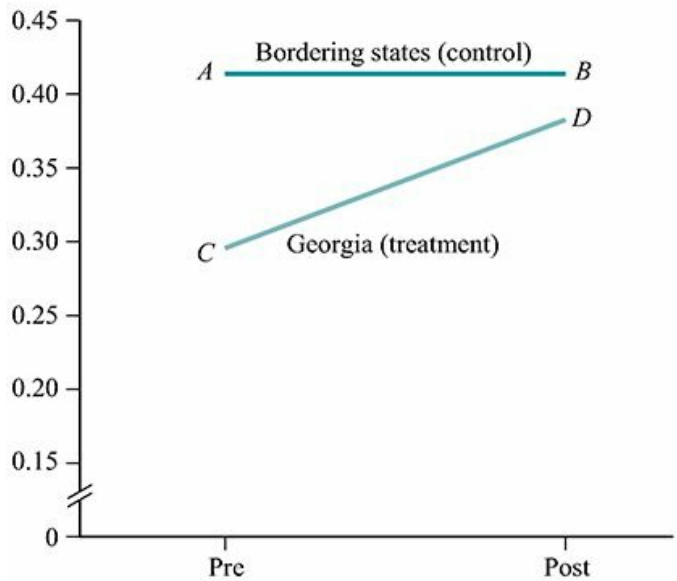

Figure 3.3
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Figure 3.3 Counterfactual Trends in Difference-in-Difference Estimation: The Case of Georgia Surrounding the Implementation of a Financial Aid Program

The difference-in-difference estimate of the effect of the Georgia HOPE scholarship on college enrollment is calculated by comparing enrollment changes in Georgia with those in bordering states

when the policy was enacted: (D – C) – (B – A).

To estimate the causal effect of the introduction of merit aid on college enrollment among high school graduates in Georgia, we calculate the change, or difference in enrollment in Georgia before and after the program introduction minus the difference in enrollment over this period in the bordering states. This *difference-in-difference estimator* takes this form:

$$\{E[Enr_{it} \mid i = GA, t = Post] - E[Enr_{it} \mid i = GA, t = Pre]\} -$$
$$\{E[Enr_{it} \mid i = Other, t = Post] - E[Enr_{it} \mid i = Other, t = Pre]\} -$$

Table 3.3 (panel A) presents this calculation schematically, which subtracts the pretreatment difference in enrollment across the treatment and control groups from the posttreatment difference.

A.

Table 3.3 The Difference-in-Difference Approach

	Treatment	Control	Difference
Pre	Enrt1	Enrt2	Enrt1–Enrt2
Post	Enrt+11	Enrt+12	Enrt+11–Enrt+12
Difference	Enrt+11–Enrt1	Enrt+12–Enrt2	(Enrt+11–Enrt1)–

			(Enrt+12−Enrt2)

In panel B we show these calculations from the analysis in Dynarski (2000) for the case of the introduction of the Georgia HOPE program:

B.

	Before 1993	1993 and after	Difference
Georgia	0.300	0.378	0.078
Rest of southern states	0.415	0.414	−0.001
Difference	0.115	0.036	**0.079**

Data from: Dynarski (2000).

College enrollment grows by 7.8 percentage points in Georgia, while the change over this interval in the comparison states is slightly negative. The result is an estimated increase in college enrollment attributable to the HOPE program (i.e., a difference-in-difference estimate) of about 7.9 percentage points.

Difference-in-difference estimators have significant appeal for education analysts and have been used widely in a number of research settings. The validity of these estimates rests on the assumption that the policy change of interest is the only reason why outcomes among the treated group change relative to outcomes among the control group.

Difference-in-Difference in Regression Form

While difference-in-difference models can be represented simply in tabular form, as shown in Table 3.3, the most common way to estimate a difference-in-difference model is using regression analysis. Let's say we have data on college enrollment for each year and state in the South from 1990–1998. For the Georgia merit aid analysis, we can represent the model as follows:

$$Enroll_{it} = \alpha + \gamma Georgia_{it} + \upsilon Post_t + \delta Georgia_{it} \times Post_t + \varepsilon_{it}$$

where i indexes the state and t indexes the year.[7]

The variable *Georgia* is an indicator, or **dummy variable**, that is equal to 1 if the state is Georgia and is equal to zero otherwise. It controls for any fixed, unchanging differences between Georgia and the control states. The *Post* variable is a dummy equal to 1 after the implementation of the merit aid program (in 1993) and equal to zero before. It controls for any systematic changes occurring across all states after 1993 that have nothing to do with the Georgia program (like an economic shock). *Post* does not vary across states: It refers to whether the year is 1993 (the first year of the Georgia program) or later. The interaction of *Georgia* and *Post* gives the difference-in-difference estimate, which is δ in this model. It shows how enrollment changes in Georgia relative

to other southern states when the program is implemented. In its current form, δ will be *identical* to the estimate in panel B of Table 3.3. It is easy to adapt this model to control for time-varying observable characteristics (such as student income and racial composition) of each state as well as in the regression. This will cause the estimate of δ to diverge from the estimate in Table 3.3.

dummy variable

A binary variable that takes on a value of 1 if a condition is met and zero if not. For example, the variable Georgia takes on a value of 1 if the observation is for Georgia and a value of zero if the observation is for any other state.

Fixed Effects

It is common that we observe individuals, schools, or states at multiple points in time in a given dataset. Such datasets are examples of *panel data*, which are defined by multiple observations of the same unit over time. With panel data, we can account for fixed differences between individuals, states or other units of observation that relate to outcomes. Florida and Virginia differ in a number of dimensions that are unlikely to change much over time, like climate and demographics; Mike and Sarah differ in lots of ways, such as height, hometown, and favorite TV shows. But, we need not observe or quantify all of these fixed differences between

states or people to control for these factors in a regression context. Instead, we can simply include a dummy variable or a *fixed effect* for each unit in our regression. This approach allows each unit to have its own intercept, which means that it accounts for all characteristics of the unit, both observed and unobserved, that are unchanging over time.

> **Quick Hint:** When we have a panel data set with many units (such as people or schools), it is possible that there will be thousands of fixed effects. A nice statistical result is that instead of controlling for lots of unit fixed effects, one can instead control for fixed differences across units of observation by subtracting out the unit-specific mean of each variable for every observation. De-meaning the data in this way can reduce computational complexity significantly.

A model that includes fixed effects is called a **fixed effects estimator**. It provides a more general way to estimate the difference-in-difference effect of the Georgia merit aid scholarship on enrollment than the approach discussed previously. In this case, the unit of analysis is the state, and there are multiple observations of each state over time. By controlling for a set of state and year fixed effects, which entails including in the model a dummy variable for each state and each year, we can account for fixed differences over the period of our sample across each state and for fixed differences across states in each year. The

variation that remains to estimate the difference-in-difference coefficient is within-state over time. This model can be written in the following form:

fixed effects estimator

Fixed effects control for fixed differences across units of observation and across different units of time with a series of indicator variables for each unit of analysis and each unit of time. For example, a fixed effects estimator with observations of students in each year would include an indicator variable for each student and an indicator variable for each year.

$$Enroll_{it} = \alpha + \beta Georgia_{it} \times Post_t + \gamma_i + \upsilon_t + \varepsilon_{it}$$

In this model, δ still is the difference-in-difference estimate, while now we have a set of state fixed effects (γ_i) and a set of year fixed effects (υ_t). This is a more general model than the one we presented previously, because now we control for fixed differences across each state, not just across Georgia and the rest of the South. We also control for fixed characteristics of each year, not just the way in which the post period differs on average from the pre period.

While we motivated fixed effects analyses using state year data, the use of fixed effects to estimate difference-in-difference models is pervasive. One example is the use of individual student fixed effects to estimate the effect of

charter schools on educational achievement (see Chapter 10). In these models, the student fixed effects control for average fixed differences across individual students. As a result, they tell us how achievement *changes* for students who enter a charter school relative to students who remain in their traditional public school. The idea is that the students who do not change schools tell us about the counterfactual change in outcomes among students who do change schools.

Instrumental Variables

Sometimes, life events are shaped by random events: whether or not young men were drafted to serve in the Vietnam war was directly related to lottery numbers. The enrollment of a student in a charter school that is oversubscribed also depends on the outcome of a random lottery. These lotteries produce **instruments** or **instrumental variables** that can be used to disentangle the causal effects of activities like military service or charter school enrollment on outcomes like long-term earnings or academic achievement. Because winning or losing the lottery is random, comparing outcomes of winners versus losers essentially mimics a randomized controlled trial.

instrument (instrumental variable)

A variable that isolates variation in the treatment that is uncorrelated with underlying characteristics of those who

are treated or untreated.

In a seminal paper on the effect of education on earnings, Angrist and Krueger (1991) use quarter of birth as an instrument for educational attainment. We discuss this paper at length in Chapter 6, but the idea is that states have school starting ages that historically have mandated that students must be 6 in January of the year in which they enter school. This is combined with compulsory schooling laws that mandate a student must be enrolled until he or she is a certain age. As a result, a student born on January 1 will start school a year later than a student who is born the day before, on December 31. The January 1 child will therefore be a year behind in school when he reaches the compulsory schooling age. Angrist and Krueger show that, indeed, those born in the first two quarters of the year get less education than those born later in the year.

Quick Hint: Historically in the United States, schools had January 1 cutoff dates for ages of entry. In the older cohorts of the Angrist and Krueger study, this cutoff was common across states. Since that time, school age cutoff dates have changed considerably and now vary across states.

Quarter of birth acts as an instrument for education because it isolates the variation in educational attainment that is as good as random: The variation in education because of

quarter of birth is assumed to be unrelated to underlying characteristics of those who obtain more versus less education. If quarter of birth is essentially randomly assigned, we can compare differences in earnings across people whose education levels differ only because they have different quarters of birth. This will isolate the causal effect of education on earnings. In essence, an instrument pulls out the part of the variation that is akin to a randomized controlled trial.

Instrumental variables (or IV) estimation in this setting works by isolating the variation in education *that is only due to when in the year you were born* to estimate the effect of schooling on earnings. This works under two assumptions:

The instrument must be correlated with treatment: Quarter of birth must actually affect educational attainment.
The *only* reason the instrument and the outcome are correlated is because the instrument affects the treatment. This means that the only reason quarter of birth is related to earnings is because it affects educational attainment.

The nice aspect of IV is that the first assumption can be tested with the data very easily. The drawback is that the second assumption cannot be tested. This is the major source of controversy surrounding the use of instruments: It often is hard to know whether there are other reasons that the instrument is correlated with the outcome of interest.

In the education–earnings example, the question is

whether there is *any* other mechanism through which quarter of birth can influence earnings. What if people born earlier in the year are in families with lower income or have parents with less education? What if there is an effect of being older in one's class (as the early-quarter children are) on educational attainment? These are all threats to this type of design. When you see an instrumental variables strategy, it is useful to go through ways it would fail to help determine whether the assumptions underlying the instrument are plausible or not.

Many government policy changes can be used as instruments. As long as the policy change is random with respect to unobserved changes in a school or state, these policy changes can induce quasi-random variation that can be used to estimate causal effects. When we see a paper that uses quasi-experimental variation, we should subject it to the same type of scrutiny as an instrumental variables model. We will get much practice at this throughout the book as we discuss empirical results from economics research.

Regression Discontinuity: Examining Sharp Breaks

Policies that generate sharp breaks in eligibility for a treatment get a lot of use and attention among education researchers. Consider some policy rules that may generate sharp breaks, or discontinuities, in eligibility for education

programs and resources:

- Rules that mandate maximum class size at a fixed cutoff, say 30 students
- Regulations stipulating the minimum age of school entry at a specific date, such as age 5 by September 1
- Policies that designate eligibility for financial aid at a particular achievement score cutoff or income threshold
- University admission policies that admit all students with an SAT score above a certain mark

If we are willing to assume that students or schools just above and below the cutoff in these cases are nearly identical in observed and unobserved characteristics, then the comparison of student or school outcomes just above and below the threshold provides a good measure of the causal impact of the education policy on such outcomes. This sharp break can be thought of as an instrumental variable: It generates variation in the treatment that is unrelated to underlying characteristics of students or schools.

Suppose there is a $5,000 college scholarship with a strict test score threshold for qualification. Because there is some random noise in test scores, we would not expect to see a difference in collegiate outcomes (such as enrollment or completion) across those right above versus right below the threshold absent the scholarship rule. The cutoff (or discontinuity) for scholarship eligibility provides an opportunity to estimate the causal effect of the scholarship on

collegiate attainment. If no one below the cutoff received the scholarship and everyone above the cutoff received the scholarship, the causal effect of the scholarship on graduation can be estimated by comparing graduation rates right above the cutoff to those right below the cutoff. The main assumption with this approach is that students are unable to manipulate their test performance to place themselves over the cutoff. That is, those just above and just below the treatment threshold are essentially the same on average.

Often, there is incomplete take-up or there are other eligibility criteria that determine treatment. The top panel of Figure 3.4 shows such a scenario. Test scores are shown relative to the test score cutoff on the x-axis; they are referred to in this scenario as the **running variable (forcing variable)**. The y-axis shows whether an individual received the scholarship: Some below the cutoff receive the scholarship and some above do not accept it. Nevertheless, there is a clear discontinuity around the cutoff of 75 percentage points. This is an example of a **fuzzy regression discontinuity design**, which occurs when the likelihood of treatment changes by less than 100 percentage points at the threshold.

running variable (forcing variable)

The variable that describes how close an individual is to a treatment threshold in a regression discontinuity design.

fuzzy regression discontinuity design

A regression discontinuity design in which the likelihood of treatment changes by less than 1 at the threshold.

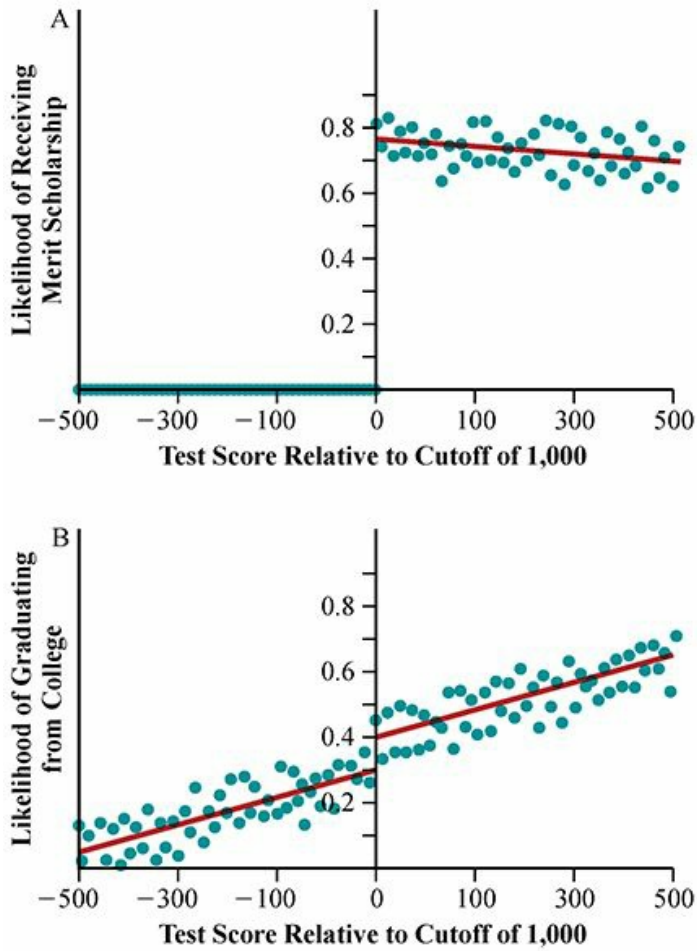

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Figure 3.4 Regression Discontinuity Illustration

A. The effect of the merit aid eligibility rule on receipt of scholarships. There is a discontinuity in the likelihood of receiving a merit scholarship at a test score of 1,000. This is a fuzzy regression discontinuity because the likelihood of receiving the

scholarship increases by less than 1 at the cutoff; some students who are eligible do not attend college in-state. The discontinuity increases the likelihood that one receives a merit scholarship by 75 percentage points. **B.** The effect of the merit aid eligibility rule on college graduation. The likelihood of college graduation jumps by 10 percentage points at the cutoff. If we assume that this increase is completely driven by the 75% of people who receive the scholarship, the effect of receiving a scholarship on college completion is 13.3 percentage points (0.1/0.75).

In the bottom panel of Figure 3.4, there is a corresponding jump in the likelihood of college completion at the cutoff of 10 percentage points. If this increase is determined by the increase in scholarship receipt, the effect of receiving a scholarship on college completion is 13.3 percentage points (0.1/0.75). The discontinuity in the assignment rule thus acts as an instrument for scholarship receipt: The instrumental variable estimate of the effect of scholarship receipt on graduation is 13.3 percentage points.[8]

The main assumption underlying this method is that the students just above and just below the break are very similar. One way to think about this condition is that there must be some randomness in test outcomes, such that one cannot determine on which side of the discontinuity one locates. As long as this randomness exists in an area local to the cutoff,

then those just above and below the cutoff will be the same on average (since they are allocated by the random test outcomes). Running variables that have randomness associated with them, like test scores, are often used in these settings. Forcing variables without this property, like income, can cause more problems, as people typically can bunch right above the cutoff. Thus, either it must be difficult to adjust around a known cutoff (the case of birthdays or test scores) or the cutoff must be unknown to individuals. As long as one cannot choose what side of the discontinuity to be on, we can interpret the magnitude of the change around this break as a causal estimate of the effect of financial aid on college completion.

Regression discontinuity estimates of program effects are specific to the region of the cutoff—what economists refer to as local treatment effects. Regression discontinuity designs can tell us about causal treatment effects for those who are right next to the cutoff. It cannot tell us how those far away from the cutoff are affected. This is an important caveat to these studies, as often we use test scores to determine eligibility. We cannot learn how very high- or very low-achieving students are affected by the treatment using regression discontinuity methods; we only know the impact among those whose test scores place them right near a cutoff. This is a key limitation of regression discontinuity studies that affects the external validity of the findings.

3.4 Conclusion

How we establish a causal relationship between a treatment and an outcome is a core question in the economics of education. Economic theory is critically important in generating hypotheses about how policy innovations ranging from class size reduction to changing the age of compulsory school attendance to increasing financial aid affect outcomes. To inform policy, it is imperative to test the predictions of theory and evaluate education policies. Evaluating education policies is challenging because we need to move beyond correlational measures to capture the causal impact of education on student and school outcomes. In the next chapters, we present both theoretical models of how we expect education policies to affect these outcomes and empirical evidence on the causal effects of education-related policies.

Highlights

To study the effectiveness of an education policy, it is essential to estimate a **causal link**, or the **treatment effect** of the policy, not just a **correlation** among variables. There

also can be **heterogeneous treatment effects**, whereby the treatment effect differs across different types of individuals. We use the tools of **econometrics** to help us determine whether (and for whom) a given policy is causing a specific outcome or whether it is simply correlated with that outcome.

The idea of the **counterfactual** is essential to assessing causation. The counterfactual is the outcome that would have occurred absent the treatment. Because we see people in only one state of the world (treated or untreated) at any given time, the counterfactual is never observed. We must find other groups—control groups—that allow us to measure the counterfactual.

Obtaining an accurate measure of the counterfactual is complicated by **selection**: People can choose whether to receive the treatment, so those who select into the treatment will be systematically different from those who do not. Comparisons among the treated and untreated groups will reflect both the difference in their characteristics and the effect of the treatment.

Empirical results have **internal validity** if the estimated causal effect is unbiased, that is, if we are not using a **biased estimator**. They have **external validity** if the results can be generalized to other individuals or settings.

Randomized controlled trials (RCTs) are the most straightforward way to estimate the counterfactual in a manner that overcomes selection bias. Because people are

randomly assigned to receive the treatment or not, the treatment and control groups are, on average, identical. RCTs are the gold standard for policy analysis, as the randomization overcomes any selection bias and the treatment effect is easily calculated by comparing differences in mean outcomes across the treated and untreated groups.

Experiments in education are difficult to conduct. Such experiments face several obstacles, including **Hawthorne effects**, necessary time lags to observe outcomes of interest, high costs, the inability to compel people to comply with their assignment, and ethical boundaries. As a result, we need to develop nonexperimental methods that allow us to overcome *selection bias*.

Economists studying education often rely upon quasi-experimental methods, which mimic experiments by using naturally occurring variation in treatments.

Multivariate regression, which entails one **dependent variable** and multiple **independent (explanatory) variables**, is a powerful tool that economists can use to overcome selection bias in many circumstances. We can include control variables in a regression that account for differences between the treatment and control group, but we are limited by the variables in the data. If there are **omitted variables** in the regression model that are correlated both with the treatment and with the outcome, it will create **omitted variables bias** in the estimates. A **natural**

experiment (quasi-experiment) that generates variation in the treatment unrelated to differences between the treated and untreated groups also can be used to address selection bias. All nonexperimental papers discussed in this text use one of these methods to estimate causal effects.

When there are changes in education policies or laws, one often can overcome **selection bias** by comparing changes in outcomes among the treated group to changes among the control (or unaffected) group. This difference-in-difference method often is implemented using a **fixed effects estimator** that entails controlling for a series of **dummy variables** that account for fixed differences across groups.

Instrumental variables (IV) can be used to isolate variation in the treatment assignment that is uncorrelated with the underlying characteristics of those who are at risk of treatment. The main assumptions underlying instrumental variables are that the **instrument** is strongly correlated with the treatment and that the *only* reason the instrument is related to the outcome is because of its relationship with the treatment. While a powerful tool, the second assumption cannot be tested, which leads to much controversy and discussion related to instrumental variables studies.

Regression discontinuity methods also are widely employed in economics. This method is used when treatment status changes abruptly (i.e., discontinuously) based on some characteristic of an individual or school. A **fuzzy regression discontinuity design** is one in which the

likelihood of treatment changes by less than 1 at the cutoff. Rules that determine college admissions or eligibility for a gifted and talented program based on test score cutoffs, for example, allow for a regression discontinuity analysis. This method works as long as people cannot choose which side of the discontinuity to be on, which usually requires some randomness in the **running variable (forcing variable)** that assigns treatment. In such cases, we can estimate the causal effect of a program by comparing people just above with those just below the cutoff.

Problems

What is selection? Why does self-selection make it difficult to measure the effect of education policies?

Consider the following scenarios and explain why the correlation described need not imply that one factor causes the other. Be explicit about the other factors that might be at play.

a. Students in charter schools have higher test scores than their counterparts in nearby public schools.
b. A negative correlation between maternal smoking and their children's birth weight.
c. On average, students with relatively large student loans are less likely to default.
d. Within a school, students in smaller classes have lower

end of year test scores.

e. Children in families that spend more time on educational activities do not have better educational outcomes than children in families who spend less time on educational activities.

Explain how randomized controlled trials overcome selection bias.

What is the difference between internal validity and external validity? Does the fact that a study is internally valid mean it is externally valid?

What problems arise in Randomized Control Trials (RCTs) in which treatment assignment is not double-blind? How might these problems bias the estimates of an experiment that randomly assigns some schools in a city to receive more financial resources?

Some financial aid such as the Pell grant is assigned on the basis of financial need. Suppose you examine the relationship between college completion and financial aid receipt by running a regression of college completion on aid receipt. Why might it be incorrect to interpret the coefficient on financial aid eligibility as the causal effect of aid on college completion?

In 2001, the federal government enacted a law that forbade any student with a drug offense from receiving financial aid. Lovenheim and Owens (2014) study the effect of this law on college enrollment behavior by examining how enrollment among those with convictions changes relative

to those without convictions in a difference-in-difference analysis. A difference-in-difference analysis takes two successive differences. The first difference is the difference between the pre and post treatment values of a group that underwent treatment and the difference between the pre and post treatment values of area group that did not undergo treatment. Once these two values are calculated, the difference between these differences is computed.

a. Using the provided information, calculate the difference-in-difference estimate.

	No Conviction	**Conviction**
Pre-change	0.623	0.358
Post-change	0.652	0.269

b. Does the fact that college enrollment was rising throughout the United States for all students over the period cause a problem for their study?
c. Write the regression formula that will allow you to estimate the difference-in-difference treatment effect.
d. How might selection bias the results?

Regression discontinuity can be a useful strategy for estimating the causal effect of a policy with a sharp break in eligibility for treatment. What are the two main considerations of this approach?

Gifted and talented programs provide extra services for academically high-performing students. Eligibility for these programs is typically based on having a test score above a

threshold. A study by Bui, Craig, and Imberman (2014) estimate the effect on achievement of participating in a gifted and talented program using a regression discontinuity design.

a. Explain in words how the regression discontinuity method leads to a valid estimate of the causal effect of being in a gifted and talented program on educational outcomes.
b. Can this method tell us how very high-achieving students are affected by the gifted and talented program?
c. Would it be a problem for this approach if not all students who were eligible for the program decided to enroll in it?